

# Project Report: Telco Customer Churn Analysis

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### **Introduction**

This project aims to analyze and predict customer churn in a telecommunications company. Customer churn, or customer attrition, represents the loss of clients or customers. Predicting customer churn is crucial for telecommunications companies to proactively identify customers who are likely to cancel their service, allowing for targeted retention efforts.

### **Dataset Description**

Number of features: 21 (including customerID and target variable)

Problem type: Binary Classification (Churn: Yes/No)

Number of data points: 7043 customers

### **Feature types:**

Quantitative: MonthlyCharges, TotalCharges, tenure

Categorical: gender, InternetService, Contract, PaymentMethod, etc.

Categorical variable encoding: Required for all categorical variables to convert them into numeric format for model processing

### **Correlation analysis:**

Strong positive correlation between TotalCharges and tenure

Moderate correlation between MonthlyCharges and TotalCharges

Other features show weak to moderate correlations

### **Class distribution:**

No Churn: 73.46%

Churn: 26.54%

**Therefore, the dataset is imbalanced.**

### **Dataset Pre-processing**

#### **Identified Issues:**

Missing Values

TotalCharges column contains some missing values

Solution: Removed rows with missing values due to small number of affected rows

Categorical Values

Multiple categorical variables present

**Solution:** Label Encoding applied to convert all categorical variables to numeric format

Feature Scaling

Solution: StandardScaler applied to normalize all features

### **Dataset Splitting**

Training set (80%): 5634 samples

Testing set (20%): 1409 samples

Stratified splitting used to maintain churn distribution in both sets

**For Model Development and Evaluation, three different models were implemented and compared:**

### **K-Nearest Neighbors (KNN)**

#### **Decision Tree**

#### **Neural Network**

#### **Model Performance Analysis**

### **K-Nearest Neighbors (KNN)**

Accuracy: 74.84%

Precision: 74.88%

Recall: 74.84%

F1-Score: 74.86%

ROC-AUC: 77.08% The KNN model showed solid overall performance with balanced metrics and the highest ROC-AUC score among all models.

### **Decision Tree**

Accuracy: 71.07%

Precision: 71.95%

Recall: 71.07%

F1-Score: 71.47%

ROC-AUC: 64.52% The Decision Tree model showed the lowest performance across all metrics, particularly in ROC-AUC score.

### **Neural Network**

Accuracy: 76.33%

Precision: 74.76%

Recall: 76.33%

F1-Score: 75.16%

ROC-AUC: 76.87% The Neural Network achieved the highest accuracy and F1-Score among all models, with strong performance across all metrics.

## Model Comparison and Selection Best Performing Model:

### **Neural Network**

Highest accuracy at 76.33%

Most balanced performance across all metrics

Strong ROC-AUC score of 76.87%

### **Second Best: KNN**

Slightly lower accuracy but more consistent precision and recall

Highest ROC-AUC score at 77.08%

### **Poorest Performance: Decision Tree**

Lowest performance across most metrics

Significantly lower ROC-AUC score at 64.52%

Conclusions and Recommendations

### **Model Selection:**

The Neural Network model is recommended for implementation due to its superior overall performance and balanced metrics.

### **Conclusion**

#### **Performance Insights:**

All models achieved accuracy above 70%, indicating reasonable predictive power

The neural network's higher accuracy suggests better capability in handling complex patterns in the data

The decision tree's lower performance might indicate that the churn patterns are not easily separable by simple decision boundaries

#### **Future Improvements:**

Feature engineering to create more informative predictors

Hyperparameter tuning for all models

Ensemble methods could be explored to improve performance

Addressing class imbalance through techniques like SMOTE or class weights

Collecting additional relevant features that might influence customer churn

#### **Business Implementation:**

Deploy the Neural Network model for churn prediction

Implement regular model retraining to maintain performance

Establish monitoring systems for model performance

Create an automated alert system for high-risk customers

Develop targeted retention strategies based on model predictions

This concludes the analysis of the Telco Customer Churn prediction project, with the Neural Network model recommended as the primary solution for implementation.

